

How AI Answers

In Vector Space, you watched a token's final vector land in its neighborhood on the map. That landing was about meaning: the model understanding the words you put in. Now we move to the next step: how AI answers you.

Start with your phone. As you type a text, it suggests the next word: three choices above the keyboard, picked from the last word or two. And everyone gets the same suggestions.

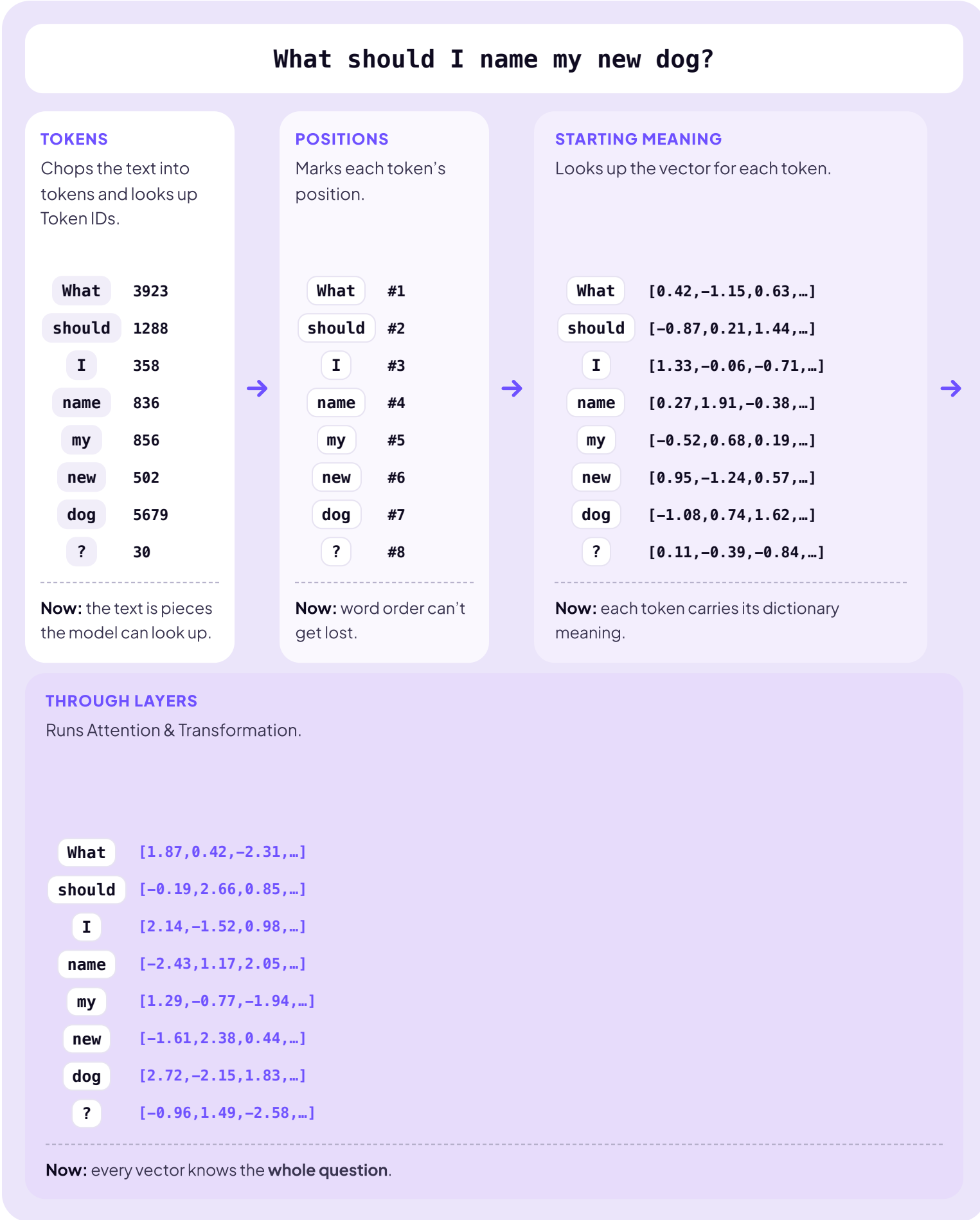
See you _____ .

soon

tomorrow

later

AI follows the same basic logic as your phone's suggestions, but, as you might expect, it goes **way deeper**. Let's take a simple question and watch AI build its answer.



WHERE WE STAND

Look at what the model has now. The question isn't words anymore. It's eight rich vectors, each one carrying what its token means inside this exact question. **Meaning: established.**

Now the detail everything turns on: **the last token matters most**. Attention doesn't just sharpen each token in place. Every pass also folds the earlier tokens into the later ones, so the last token's vector soaks up the entire question. By the final layer, the last token isn't really about the question mark (?) anymore. Attention and transformation have pushed its vector across the map, and it lands in the neighborhood of the words most likely to come next: the first word of the answer.

THE ANSWER, WORD BY WORD

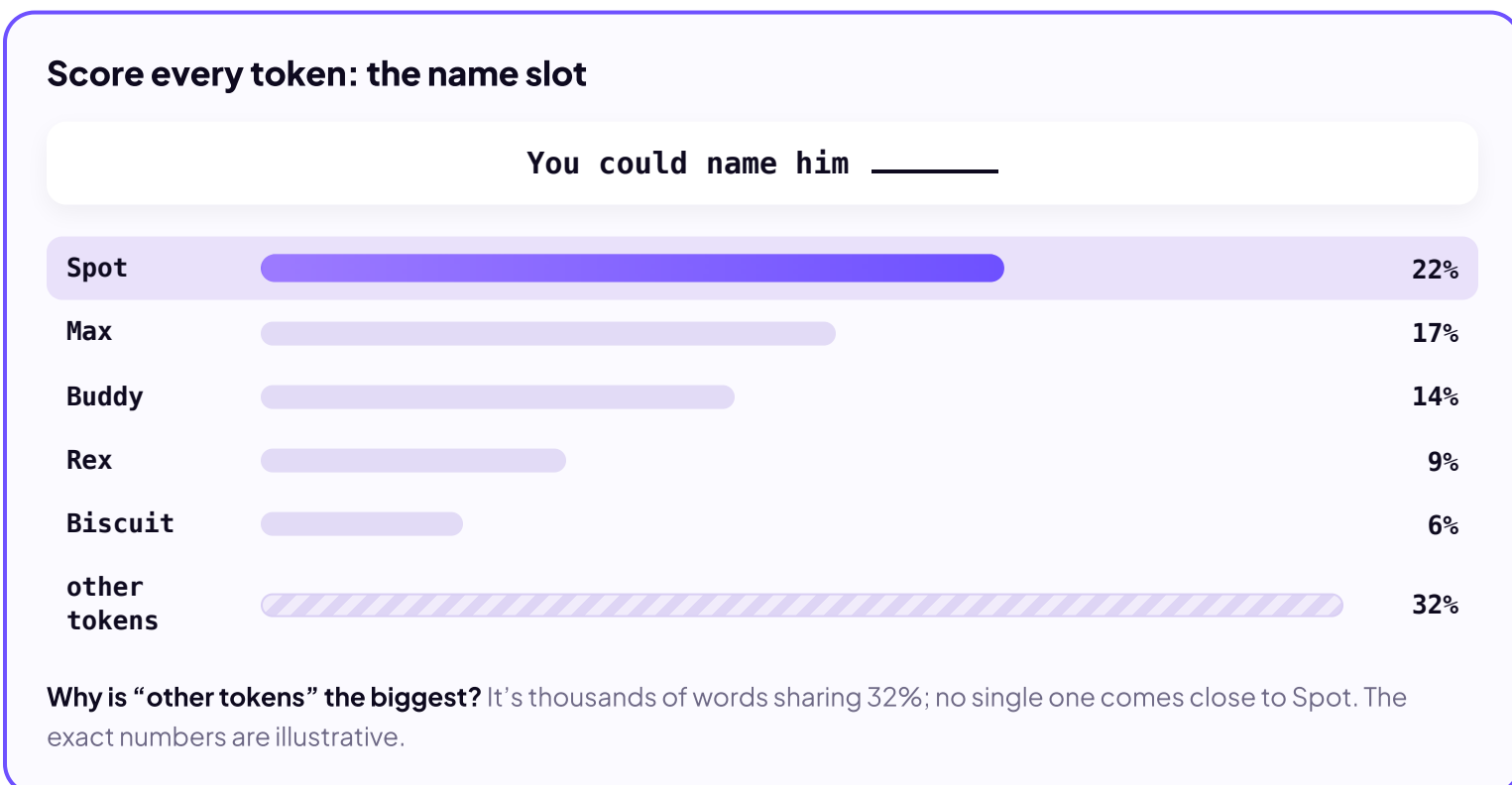
Now watch it happen. Each row below is one word of the answer. The model looks at the last token (the first column), checks which neighborhood its vector landed in, and takes the best-scoring word there. That word gets typed, joins the context, and becomes the new last token. Then the next row repeats the move. That move is called **prediction**.

What should I name my new dog?			
LAST TOKEN	REPLY SO FAR	NEIGHBORHOOD	TOP PREDICTIONS
?	—	Reply starters	You 18% A 14% Great 9%
You	You —	Ways to suggest	could 24% can 12% should 10%
could	You could —	Naming verbs	name 31% call 19% try 7%
name	You could name —	Who gets named	him 45% your 21% the 8%
him	You could name him ...	Dog names	Spot 22% Max 17% Buddy 14%
✔ You could name him Spot.			

You asked for a name for your new dog. But notice the name never came first. The neighborhoods tell the story: the sentence built its way, one token at a time, to the spot where it suggested Spot. Every row is the same move; the only thing that changes is the context it starts from.

THE RANKED LIST

One more idea to clarify. The table shows three predictions per row, but that's just the top of the list. In reality, the model assigns a probability to every token in its vocabulary, all 200,000+ of them, and ranks them all. Here's the name slot under the microscope: the moment Spot arrived.



The model takes the top of the list and types it: Spot.

And that's the whole machine. Everything you just watched, from tokens to vectors to a ranked list to the next word, has a name: **inference**. It's what the model does every time it answers you.